

# WELL COMPLETION REPORT (SYNTHETIC)

OIL-DGB-003 - Digboi-003

NWIS prototype document - not an Oil India Limited operational record

|                         |                                     |
|-------------------------|-------------------------------------|
| API Number              | OIL-ASM-000025                      |
| Field / District        | Digboi / Tinsukia                   |
| Block                   | PEL-ASSAM-01                        |
| Basin                   | Assam-Arakan                        |
| Formation               | Barail (BAR)                        |
| Spud / Completion       | 2024-03-26 / 2024-05-28             |
| Measured / TVD          | 2837 m / 2801 m                     |
| Trajectory              | Vertical; inc 2.7 deg; az 222.5 deg |
| Formation Interval      | 1440-2236 m                         |
| Lithology               | Interbedded sandstone and shale     |
| Pore Pressure Reference | 2906 psi                            |
| Fracture Gradient       | 14.51 ppg                           |
| Mud System              | PHPA Water-Based Mud                |
| NWIS Well Risk          | High (58.8/100)                     |

## Casing Program

| String            | Set Depth (m) | Purpose      |
|-------------------|---------------|--------------|
| 20 in conductor   | 80            | Conductor    |
| 13 3/8 in casing  | 709           | Surface      |
| 9 5/8 in casing   | 1645          | Intermediate |
| 7 in liner/casing | 2496          | Production   |

## Significant Historical Events and Lessons

| Date / Depth         | Event             | Severity | Evidence / mitigation / lesson                                                                                                                                                                             |
|----------------------|-------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024-04-23<br>1402 m | Fishing Operation | Critical | BHA retrieval required specialized tools. Mitigation: Review BHA failure mode before resuming drilling. Lesson: Capture retrieval outcome as institutional knowledge.                                      |
| 2024-04-28<br>1604 m | Stuck Pipe        | Critical | Rotary movement deteriorated progressively. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Use offset stuck-pipe depth to increase circulation and reaming frequency. |
| 2024-05-05<br>1891 m | Fishing Operation | Critical | Normal pipe-recovery attempts were unsuccessful. Mitigation: Use approved fishing program with controlled overpull and circulation. Lesson: Capture retrieval outcome as institutional knowledge.          |
| 2024-05-13<br>2218 m | Fishing Operation | Critical | Normal pipe-recovery attempts were unsuccessful. Mitigation: Use approved fishing program with controlled overpull and circulation. Lesson: Capture retrieval outcome as institutional knowledge.          |
| 2024-04-05<br>610 m  | Mud Loss          | High     | Partial returns observed at shale shaker. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.      |
| 2024-04-10<br>803 m  | Kick              | High     | Drilling break followed by positive flow check. Mitigation: Re-establish overbalance and monitor pit/flow balance. Lesson: Escalate connection-gas trends before entering correlated pressure zone.        |
| 2024-05-14<br>2282 m | Cementing Issue   | High     | Observed returns did not match calculated displacement. Mitigation: Use staged cementing response when returns deteriorate. Lesson: Plan conservative displacement rate in weak formations.                |

## Completion Summary

The synthetic well reached 2837 m MD in the Barail interval. The NWIS knowledge record links 15 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.